

Cambridge IGCSE™

DESIGN AND TECHNOLOGY**0445/52**

Paper 5 Graphic Products

May/June 2026**MARK SCHEME**Maximum Mark: 50

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **8** printed pages.

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: - the specific content defined in the mark scheme - the specific skills defined in the mark scheme or in the level descriptions - the standard of response required as exemplified by the standardisation scripts.
2	Examiners must award whole marks (not half marks, or other fractions).
3	Examiners must award marks positively . This means that: - marks are not deducted for errors or omissions - marks are awarded for correct/valid answers as defined in the mark scheme and also for other valid answers which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate - the quality of spelling, punctuation and grammar are judged only when the mark scheme indicates that these features are specifically assessed in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must consistently apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the full range of marks defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and must not award marks with grade thresholds or grade descriptions in mind.

Annotations guidance for centres

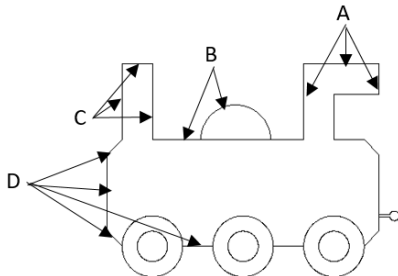
Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

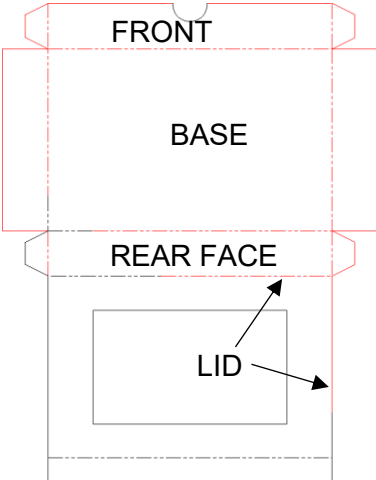
We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

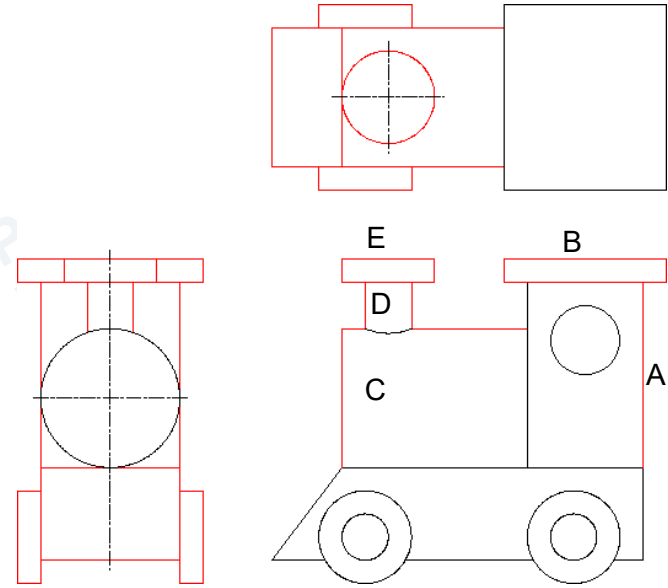
The annotations listed below were available to examiners marking this component in this series.

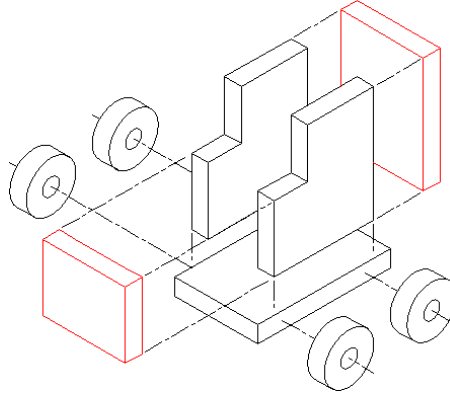
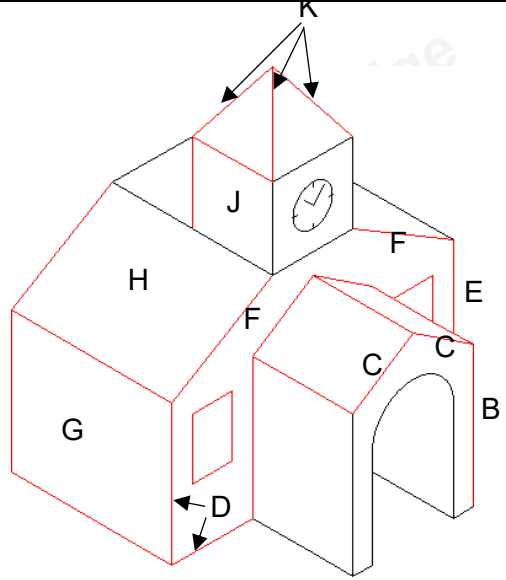
Annotations

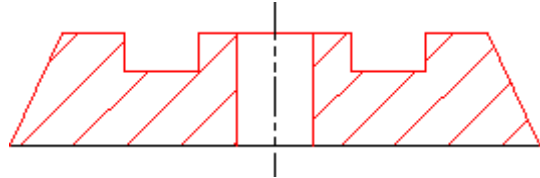
Annotation	Meaning
	Incorrect point
	Indicates that the point has been noted, but no credit has been given
	Correct point
Numbers	Indicating the mark allocated for the response

Question	Answer	Marks	Guidance
A1(a)(i)	3 lines completing the drivers cab (A) correct to overlay [1] Horizontal line for top of boiler and semi-circle (B) correct to overlay [1] 3 lines for chimney (C) correct to overlay [1] 3 lines to front of engine and baseline (D) correct to overlay [1]	4	
A1(a)(ii)	Inner circle of rear wheel $\varnothing 10$ [1] Two outer circles of rear wheels $\varnothing 20$ [1]	2	
A1(a)(iii)	Rectangle 70 mm wide or 35 mm high [1] Rectangle correct to overlay [1]	2	
A1(a)(iv)	Any isosceles triangle [1] Isosceles triangle correct to overlay [1]	2	
A1(a)(v)	Any hexagon (5 sides shown) [1] Any regular hexagon [1] Hexagon correct to overlay [1]	3	
A1(b)	Rectangle width 190 or height 105 [1] Rectangle correct to overlay [1]	2	

Question	Answer	Marks	Guidance
A2	3 Lines added at 90°, 180° and 270° [1] Lines added at 30°, 150°, 210° 330° [1] Lines added at 60°, 120°, 240° and 300° [1]	3	
A3	Lid of box and rear face correct to overlay [1] Base of box correct to overlay [1] Front Face of box correct to overlay [1] Left side face correct to overlay / candidate solution [1] Right side face correct to overlay / candidate solution [1] 4 glue flaps added in correct positions [1] Correct use of fold lines [1]	7	 The diagram shows a net for a box. It consists of several rectangular faces connected by lines representing fold lines and glue flaps. The faces are labeled: 'FRONT' at the top, 'BASE' in the middle, 'REAR FACE' below the base, and 'LID' at the bottom. Arrows point from the labels to their respective faces. The net is drawn with red lines for the main faces and black lines for the glue flaps and fold lines.

Question	Answer	Marks	Guidance
B4(a)	Side view: Vertical rear line 'A' and rectangular cabin roof 'B' correct to overlay [1] Outline of boiler 'C' correct to overlay [1] Bottom section of chimney 'D' correct to overlay [1] Top rectangle of chimney 'E' correct to overlay [1] Front view: Rectangular outline of base section correct to overlay [1] Rectangular outline of cabin correct to overlay / candidate solution [1] Roof correct to overlay / candidate solution [1] Chimney correct to overlay / candidate solution [1] Both wheels correct to overlay / candidate solution [1] Plan: Outer rectangle correct to overlay / candidate solution [1] Top edge of boiler correct to overlay / candidate solution [1] Chimney correct to overlay [1] Both wheels correct to overlay / candidate solution [1]	13	
B4(b)	Any ellipse attempted (curved / oval shape added) [1] Major axis 100 [1] Minor axis 50 [1] Up to 4 points plotted correctly [1] Up to 8 points plotted correctly [1] Ellipse correct to overlay / accurate ellipse to candidate solution [1]	6	

Question	Answer	Marks	Guidance
B4(c)	Front: 25×20 [1] 5mm thickness [1] In correct position [1] Back: 35×25 [1] 5 mm thickness [1] In correct position [1]	6	
B5(a)	Porch: Vertical side of porch 'B' correct to overlay [1] Two sloping edges 'C' to front of porch [1] 3 Porch edges projected 25 mm [1] Back edge of porch correct to overlay / candidate solution [1] Main building: Base line and vertical edge of left side 'D' correct to overlay [1] Left window added correct to overlay [1] Right side 'E' correct to overlay [1] Right window (2 lines) added correct to overlay [1] Front face of roof 'F' correct to overlay / candidate solution [1] Side of building 'G' correct to overlay / candidate solution [1] Sloping side 'H' correct to overlay / candidate solution [1] Side 'J' of clock correct to overlay [1] 3 lines 'K' of clock roof correct to overlay / candidate solution – must be correct height [1]	13	

Question	Answer	Marks	Guidance
B5(b)(i)	Offset / Lithography, flexography, digital printing, LED UV printing, gravure / roto gravure, laser, nanographic [1] Or AOVR	1	Do not allow: screen printing, large format, 3(d) , sublimation, letterpress, transfer, inkjet
B5(b)(ii)	Die cutting [1]	1	Do not allow: box cutter, laser cutter
B5(c)	Hacksaw, coping saw, band saw, hot wire cutter or AOVR [1] PVA adhesive, double sided tape or AOVR [1]	2	Allow other types of suitable saw. Allow named type of knife with long enough blade (40mm) Do not allow knives with small blades eg. Scalpel, craft knife, Stanley knife, exacto knife or 'knife' on its own. Do not accept trade names e.g. UHU, Gorilla etc.
B5(d)	Outline (height and both sloping edges) correct to overlay [1] Both rail slots correct to overlay [1] Central hole correct to overlay [1] Hatching added correctly [1]	4	
B5(e)	Lower bed / platen [1] YES and NO added and feedback arrow [1] Turn on vacuum [1] Unclamp / remove plastic sheet [1]	4	